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ACME PETROCHEMICAL DEMO COMPLEX - UNIT 2300

PROCESS FLOW DIAGRAM (PFD): CONCENTRATION & PREFLASH SECTION

Drawing Number: PFD-23-0001 | Revision: Z1 | Process Unit: Unit 2300

1. PROCESS TOPOLOGY OVERVIEW

- Dilute CHP oxidation product (Stream S210, 28 wt% CHP) enters Preflash Column V-2301 operating under 35 mmHgA c
- Overhead Cumene vapor (Stream S220, 28,000 kg/hr) is condensed in X-2301 and recycled to Oxidation.
- Concentrated 82 wt% CHP bottoms (Stream S229, 14,500 kg/hr at 92 degC) is pumped by Preflash Bottoms Pump P-230

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PFD-23-0001 PAGE 2 OF 2 - HEAT & MATERIAL BALANCE TABLE (CONCENTRATION SECTION)

Stream ID	From -> To	Fluid Service	Mass Flow (kg/hr)	Temp (degC)	Pressure	CHP wt%
S210	Oxidation -> V-2301	Dilute CHP Feed	42,500	78.0	2.4 kg/cm2g	28.0%
S220	V-2301 -> X-2301	Overhead Cumene Vapor	28,000	68.0	35 mmHgA	0.02%
S229	V-2301 -> P-2301 -> D-2304	Concentrated CHP Bottoms	14,500	92.0	6.8 kg/cm2g	82.0%